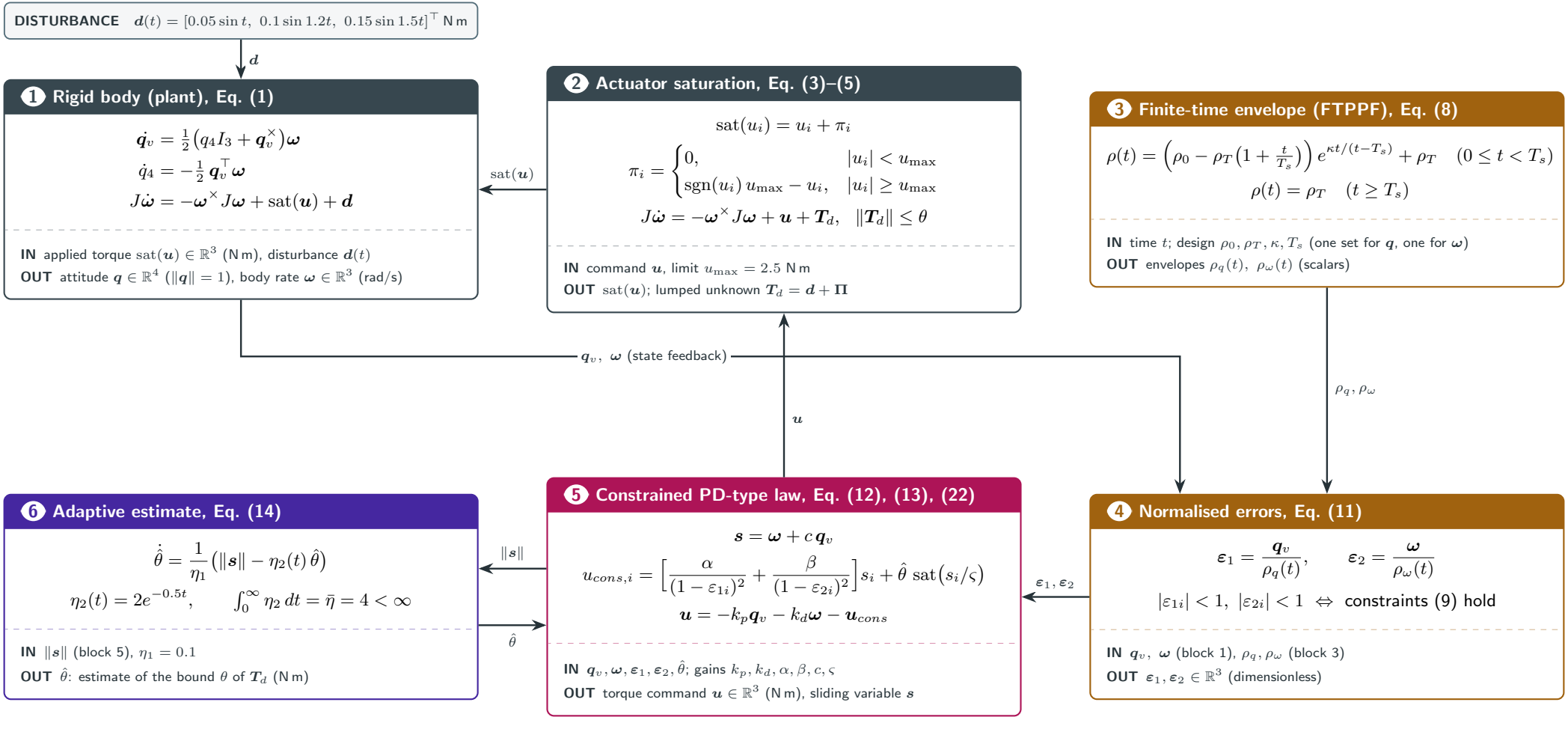

DISTURBANCE $\boldsymbol{d}(t) = [0.05\sin t,\ 0.1\sin 1.2t,\ 0.15\sin 1.5t]^\top$ N m
$\boldsymbol{d}$
1 Rigid body (plant), Eq. (1)
$\dot{\boldsymbol{q}}_v = \frac{1}{2}(q_4 I_3 + \boldsymbol{q}_v^\times)\boldsymbol{\omega}$
$\dot{q}_4 = -\frac{1}{2}\boldsymbol{q}_v^\top \boldsymbol{\omega}$
$J\dot{\boldsymbol{\omega}} = -\boldsymbol{\omega}^\times J\boldsymbol{\omega} + \mathrm{sat}(\boldsymbol{u}) + \boldsymbol{d}$
IN applied torque $\mathrm{sat}(\boldsymbol{u}) \in \mathbb{R}^3$ (N m), disturbance $\boldsymbol{d}(t)$
OUT attitude $\boldsymbol{q} \in \mathbb{R}^4$ ($\|\boldsymbol{q}\| = 1$), body rate $\boldsymbol{\omega} \in \mathbb{R}^3$ (rad/s)
$\mathrm{sat}(\boldsymbol{u})$
2 Actuator saturation, Eq. (3)–(5)
$\mathrm{sat}(u_i) = u_i + \pi_i$
$\pi_i = \begin{cases} 0, & |u_i| < u_{\max} \\ \mathrm{sgn}(u_i)\,u_{\max} - u_i, & |u_i| \geq u_{\max} \end{cases}$
$J\dot{\boldsymbol{\omega}} = -\boldsymbol{\omega}^\times J\boldsymbol{\omega} + \boldsymbol{u} + \boldsymbol{T}_d, \quad \|\boldsymbol{T}_d\| \leq \theta$
IN command $\boldsymbol{u}$, limit $u_{\max} = 2.5$ N m
OUT $\mathrm{sat}(\boldsymbol{u})$; lumped unknown $\boldsymbol{T}_d = \boldsymbol{d} + \boldsymbol{\Pi}$
3 Finite-time envelope (FTPPF), Eq. (8)
$\rho(t) = \left(\rho_0 - \rho_T\left(1 + \frac{t}{T_s}\right)\right) e^{\kappa t/(t-T_s)} + \rho_T \quad (0 \leq t < T_s)$
$\rho(t) = \rho_T \quad (t \geq T_s)$
IN time t; design $\rho_0, \rho_T, \kappa, T_s$ (one set for $\boldsymbol{q}$, one for $\boldsymbol{\omega}$)
OUT envelopes $\rho_q(t)$, $\rho_\omega(t)$ (scalars)
$\boldsymbol{q}_v$, $\boldsymbol{\omega}$ (state feedback)
ρ_q, ρ_ω
$\boldsymbol{u}$
6 Adaptive estimate, Eq. (14)
$\dot{\hat{\theta}} = \frac{1}{\eta_1}\left(\|\boldsymbol{s}\| - \eta_2(t)\,\hat{\theta}\right)$
$\eta_2(t) = 2e^{-0.5t}, \qquad \int_0^\infty \eta_2\,dt = \bar{\eta} = 4 < \infty$
IN $\|\boldsymbol{s}\|$ (block 5), $\eta_1 = 0.1$
OUT $\hat{\theta}$: estimate of the bound θ of $\boldsymbol{T}_d$ (N m)
$\|\boldsymbol{s}\|$
$\hat{\theta}$
5 Constrained PD-type law, Eq. (12), (13), (22)
$\boldsymbol{s} = \boldsymbol{\omega} + c\,\boldsymbol{q}_v$
$u_{cons,i} = \left[\frac{\alpha}{(1-\varepsilon_{1i})^2} + \frac{\beta}{(1-\varepsilon_{2i})^2}\right] s_i + \hat{\theta}\,\mathrm{sat}(s_i/\varsigma)$
$\boldsymbol{u} = -k_p\boldsymbol{q}_v - k_d\boldsymbol{\omega} - \boldsymbol{u}_{cons}$
IN $\boldsymbol{q}_v, \boldsymbol{\omega}, \boldsymbol{\varepsilon}_1, \boldsymbol{\varepsilon}_2, \hat{\theta}$; gains $k_p, k_d, \alpha, \beta, c, \varsigma$
OUT torque command $\boldsymbol{u} \in \mathbb{R}^3$ (N m), sliding variable $\boldsymbol{s}$
$\boldsymbol{\varepsilon}_1, \boldsymbol{\varepsilon}_2$
4 Normalised errors, Eq. (11)
$\boldsymbol{\varepsilon}_1 = \frac{\boldsymbol{q}_v}{\rho_q(t)}, \qquad \boldsymbol{\varepsilon}_2 = \frac{\boldsymbol{\omega}}{\rho_\omega(t)}$
$|\varepsilon_{1i}| < 1,\ |\varepsilon_{2i}| < 1 \ \Leftrightarrow$ constraints (9) hold
IN $\boldsymbol{q}_v$, $\boldsymbol{\omega}$ (block 1), ρ_q, ρ_ω (block 3)
OUT $\boldsymbol{\varepsilon}_1, \boldsymbol{\varepsilon}_2 \in \mathbb{R}^3$ (dimensionless)